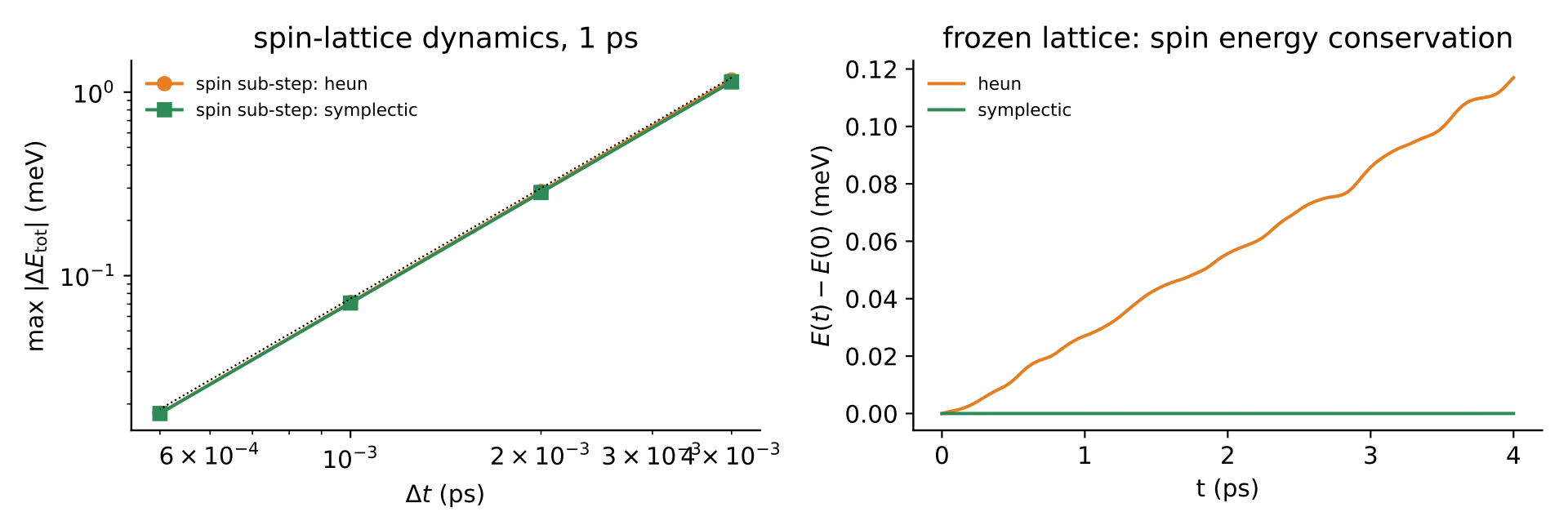
spin-lattice dynamics, 1 ps
spin sub-step: heun
spin sub-step: symplectic
max $|\Delta E_{\mathrm{tot}}|$ (meV)
10^{0}
10^{-1}
6×10^{-4}
10^{-3}
2×10^{-3}
3×10^{-3}
4×10^{-3}
Δt (ps)
frozen lattice: spin energy conservation
heun
symplectic
$E(t) - E(0)$ (meV)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0
1
2
3
4
t (ps)